

Session 1

Design Thinking & Problem Statement



Make sure it's worth building — for a real person.

Part	Topic	Time
01	Empathy: find a real pain point	7 min
02	Define: a problem statement and a “How might we” question	8 min
03	Ideate: brainstorm with the INCIPE building blocks	5 min
04	Workshop: five IoT product ideas	10 min

You'll need sticky notes or paper, a pen, your group, and the Wiki open to look up modules.

Seven words you will hear today.

Word	Plain meaning
Design thinking	Inventing by starting from people's problems, not from the technology.
User	The person who will use what you make — not you, unless you design for yourself.
Pain point	Something that annoys, worries or slows down a user.
Empathy	Understanding how someone else feels and what their day is really like.
Problem statement	One sentence: who has the problem, what they need, and why.
How might we	A question that turns a problem into an invitation to find many solutions.
IoT	Internet of Things: everyday objects with sensors, a small computer and often a network.

Listen before you prescribe.

In M3 you debugged like a doctor. Design thinking starts the same way.

A great product that solves nobody's problem is a cure for an illness nobody has.

Five stages. Today: the first three.

Stage	Question	When
1. Empathize	Who is the user, and what bothers them?	Today
2. Define	What exactly is the problem?	Today
3. Ideate	What could solve it? (Many ideas!)	Today
4. Prototype	What's the quickest way to try one?	Session 2
5. Test	Does it actually help the user?	Session 2

Four ways to find what bothers people.

Method	What you do	Example
Observe	Watch someone do the task, without helping.	Your aunt checks the soil with her finger every day.
“Tell me about the last time...”	Ask about a real moment, not opinions.	“I was away for a week.”
Ask “why?”	Keep asking why, up to five times.	Why did it die? No water. Why? Nobody was home...
Walk through their day	List what they do from morning to night.	Where do they forget, worry or wait?

“I want a watering robot” is a solution.

The pain point underneath is “my plants die when I’m away”.

Keep asking until you reach the pain — it leaves room for better solutions.

Three pain points, three different users. 3 minutes.

Area	Imagine a user like...
Plant care	A student who travels for weekend competitions
Home security	A grandparent living alone
Gaming	A younger brother who gets bored waiting

Pick one of the curriculum's areas and write three pain points, each from a different user.

Write the problem in one sentence.

[USER] NEEDS A WAY TO [NEED] BECAUSE [INSIGHT].

A student who travels for weekend competitions needs a way to keep their desk plant watered because nobody checks it while they're away.

Then flip it into a question: How might we help a student keep a plant alive while they're away for a few days?

Not too broad, not too narrow.

HMW	Verdict	Why
How might we fix gardening?	Too broad	No user, no clear need — you can't start.
How might we add a pump to a plant pot?	Too narrow	The solution is already inside the question.
How might we help a student keep a plant alive while away for a few days?	Just right	A clear user and need, and many possible answers.

Like a good exam question: “Write about history” is too broad; “Write the year 1066” too narrow; “Explain why the castle was built on a hill” gives room to think.

Four brainstorming rules.

Rule	Means
Quantity first	Aim for many ideas; judge them later.
No “but”	Build on others’ ideas with “yes, and...”.
Wild ideas welcome	A silly idea often leads to a good one.
Stay on the HMW	Every idea should answer your question.

What the INCIPE modules can sense.

Module	Gives you
Temperature & humidity	Temperature in °C and relative humidity 0–100 %
Light	Brightness from 0 (dark) to 1023 (bright)
Soil moisture	A raw soil-moisture reading
Ultrasonic distance	Distance in cm
Air quality	Air quality in ppm
Flame	Flame (infrared) intensity
Button · Joystick	Pressed or not · X and Y and the stick button
IR receiver	Signals from a remote control

What the INCIPE modules can do.

Module	Does
LED strip	Lights each pixel in a colour
Buzzer	Beeps or plays a tone
Servo	Turns to a position
Motor	Spins at a speed
Water pump	Pumps water
IR sender	Sends a remote-control signal
SD card	Saves data to files

Which input notices it? Which output acts?

HMW	Input notices...	Output does...	Idea
...keep a plant alive while away?	Soil moisture: dry	Water pump: waters it	Self-watering pot
...help a grandparent know someone is at the door?	Ultrasonic: someone close	Buzzer + LED strip	Visitor alert
...make waiting for the bus less boring?	Joystick + button	LED strip as a screen	Pocket game

Five IoT product ideas in 10 minutes.

Step	Time	Do this
1	2 min	Everyone reads out their three pain points.
2	2 min	Pick two pain points. Write a problem statement and an HMW for each.
3	4 min	Brainstorm silently, one idea per sticky note, then share. Aim for 10+.
4	2 min	Pick your best five and fill in the template.

Your five ideas.

#	Idea name	User	HMW it answers	Inputs	Outputs
1					
2					
3					
4					
5					

Check each: a real user? Every module in the building-block tables? Explainable in one sentence? Keep it — Session 2 tests one.

After your own brainstorm, ask the AI for more.

IN THE WORKSPACE AI CHAT

Here is our How might we question: [HMW]. Suggest ten product ideas that use only these INCIPE modules: [paste the two building-block tables]. One sentence each.

Treat its answers like a teammate's sticky notes: some great, some naming parts the kit doesn't have — cross those out.

“I want an app that reminds me to water my plants.”

Pain point or solution?

READ THE CODE

```
"I want an app that reminds me  
to water my plants."
```

ANSWER

Solution

WHY

The pain underneath: “I forget to water my plants and they die.” Starting there leaves room for better solutions.

What's wrong with “How might we improve home security?”

READ THE CODE

```
How might we improve  
home security?
```

ANSWER

Too broad

WHY

No user, no specific need. Better: “...help a grandparent living alone know when someone is at the door?”

What's wrong with “How might we put an ultrasonic sensor on the front door?”

READ THE CODE

```
How might we put an  
ultrasonic sensor on  
the front door?
```

ANSWER

Too narrow

WHY

The solution is already in the question, so the brainstorm has nowhere to go.

Your HMW is about knowing when a room is too hot. Which input notices it?

READ THE CODE

```
Inputs: temperature & humidity,  
light, soil moisture, ultrasonic,  
air quality, flame, button, joystick
```

ANSWER

Temp & humidity

WHY

It gives the temperature in °C.

Put the five design-thinking stages in order.

READ THE CODE

```
Test · Ideate · Empathize ·  
Prototype · Define
```

ANSWER

E → **D** → **I** → **P** → **T**

WHY

Empathize, Define, Ideate, Prototype, Test. Today covered the first three.

Three things to take away.

1

Start from people: observe, ask about real moments, and keep asking “why?” until you reach the pain point.

2

Write a problem statement, then a How might we question — not too broad, not too narrow, no solution inside.

3

Brainstorm many ideas first, then match each to the INCIPE inputs that notice the problem and the outputs that act.

Next — Session 2: Product-market fit & solution validation. Who exactly wants your idea, a Lean Canvas, and a 3-minute problem-statement pitch.